

[← Go Back](#)

Hardware

Portenta X8

Tutorials

01. Portenta X8 User Manual
02. Portenta X8 Fundamentals
03. Uploading Sketches to the M4 Core on Arduino Portenta X8
04. Data Exchange Between Python® on Linux & Arduino Sketch
05. Managing Containers with Docker on Portenta X8
06. Using FoundriesFactory® Waves Fleet Management
07. Deploy a Custom Container with Portenta X8 Manager
08. How To Build a Custom Image for Your Portenta X8
09. How To Update Your Portenta X8
10. Data Logging with MQTT, Node-RED, InfluxDB and Grafana
11. Output WebGL Content on a Screen
12. Multi-Protocol Gateway With Portenta X8 & Max Carrier
13. Running WordPress & Database Containers on Portenta X8
14. Portenta X8 Firmware

Home / Hardware / Portenta X8 / 05. Managing Containers with Docker on Portenta X8

05. Managing Containers with Docker on Portenta X8

This tutorial shows how to install and manage your containers using Docker.

Author · Pablo Marquínez · Last revision · 09/25/2024

ON THIS PAGE

Overview

Goals +

Using Docker +

Conclusion +

Overview

Docker is a platform full of applications called containers. Containers are isolated solutions; thus, they do not have to depend on your environment, making them portable and consistent throughout development, testing, and production.

You can download, install, use, and share applications in the form of containers. You can find all the available container images on the [hub.docker.com](#) page.

In this tutorial, we will go through the steps of how to install, run, and remove Docker's official [Hello-World image](#).

Goals

- Learn how to list installed and active containers
- Learn how to install a

Learn how to run a container manually

Learn how to uninstall a container

Hardware and Software Requirements

Portenta X8

USB-C® cable (USB-C® to USB-A cable)

Wi-Fi® Access Point with Internet Access

ADB: [Check how to connect to your Portenta X8](#)

Arduino IDE 1.8.10+, Arduino IDE 2, or Arduino Cloud Editor



Make sure to have the Portenta X8 with the latest image as well as the bootloader. Please check [how to flash your Portenta X8](#) to have the latest version.

Using Docker

The Portenta X8 provides Docker CLI by default. To verify its correct installation, use the following command:

```
1 docker -v
```

Or as well:

```
1 docker --version
```



To use this tool, you will need to connect to your device first. Check [how to connect using adb/ssh](#).

You can explore Docker's comprehensive reference documentation, covering all tool features in depth, at docs.docker.com.

The following steps will show how to pull the **"Hello World"** image from Docker Hub, run the container, and view its status.

To avoid a lack of permissions while launching `adb shell`, you may type the following:

```
newgrp - docker
```

The previous command and other important info about Linux on your Portenta are described in the [Portenta X8 User Manual](#).

How to Install a Container

First, you will need to search for **"Hello World" container image**. The container image can be found within the Docker hub, where you will be able to find a variety of readily-available container images. It will be used to verify docker is working as intended with the Portenta X8.

The following command is used to pull the `hello-world` image. The Docker hub page for images has the instructions to pull the image and deploy the container.

```
1 docker pull hello-world
```



Docker CLI pulling a container

Run The Installed Container

Use this command to begin a container from the `hello-world` image:

```
1 docker run hello-world
```



Docker CLI running Hello World app

To see a list of active and exited containers,



`docker ps -a` should be used after running a container at least once with

`docker run`

Listing Active Containers And Available Docker Images

The following command will display

able to run successfully. The `STATUS` message will let you know if the container is active or has finished operation depending on its purpose.

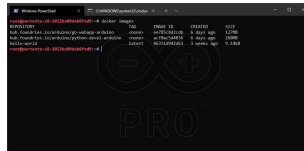
```
1 docker ps -a
```



Docker CLI listing all the active containers

The list of available images, including installed `hello-world` image, can be verified using the following command:

```
1 docker images
```



Docker CLI images

How to Remove A Container

You will need to obtain an assigned `CONTAINER ID` to remove a container of your choice. This can be found by listing all containers, including inactive ones:

```
1 docker ps -a
```

The remove (`rm`) command is then used with the desired container identifier to proceed with the removal process.

```
1 docker container rm <CONT
```

For this example, the command `docker ps -a` will show the `CONTAINER ID` of the `hello-world` container designated as: `c44ba77b65cb`.

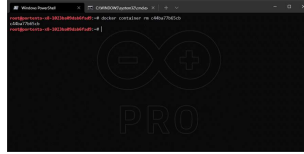
If you encounter an error stating that the container cannot be removed, it may mean that the container has an ongoing operation that can be checked with a `STATUS` message.

If this is the case, you will need to stop the container and verify with a `STATUS` message that it has exited successfully. To do this, the following command is used:

```
1 docker stop <CONTAINER ID
```

Every time the image is re-installed or re-ran, the

 `CONTAINER ID` will be different than the previous identifier



Docker CLI container uninstall

Using the `docker ps -a` after container removal, the `hello-world` container should no longer be present as an active container.

The same goes for the images if you would like to free some space. The removal command will now be as follows using `IMAGE ID` found within the image table:

```
1 docker rmi <IMAGE ID>
```

If you run `docker images` again, you will see that the `hello-world` image is not showing up anymore.

Conclusion

In this tutorial, you have learned how to use Docker with Portenta X8. You have learned to download an image, manage, and deploy the container, and ultimately run on Portenta X8. You have also learned to remove the image and the container to control Portenta X8's available resources.

Next Steps

Now that you have the base of the workflow to use [Docker](#)

sure you understand all the features.

Look for a container image from [Docker hub](#), install it and make your own application out of it.

Create a container to run your custom-made application. For this, it may interest you [Deploy a Custom Container with Portenta X8 Manager](#) tutorial.

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